

Econ 139: Labor Economics

Perfect Competition

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Perfect Competition

- Last class, we discussed a number of big trends in labor markets
- All of these depend crucially on the wage level
 - Labor share, wage inequality, labor force participation rate all depend on the wage level
- Perfect competition is a simple baseline model that will allow us to understand how changes in the labor market may impact employment and wage levels

Definition of Perfect Competition

- You probably covered perfect competition extensively in introductory economics courses, but it will be helpful to review how this works in labor markets
- To illustrate the relevant concepts, we will assume that we are a café trying to hire baristas in a perfectly competitive labor market

Assumptions of Perfect Competition

- For labor markets, the perfectly competitive model assumes that firms are wage takers
- This implies that as a café, I simply pay the market wage of baristas. If I pay a cent less than the market wage, then all of my workers will immediately leave and go to another firm
 - This assumption can be rationalized if there are a large number of competitors and the costs of moving jobs is zero
 - Does this sound realistic to you?

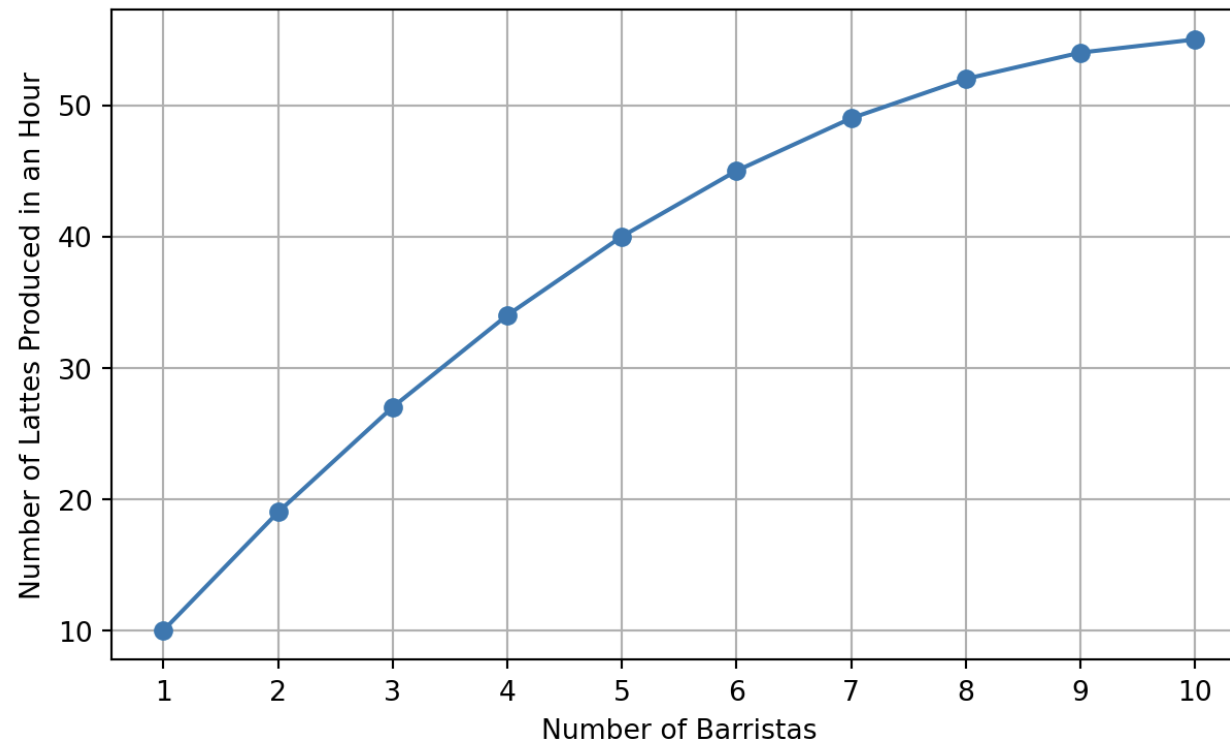
The Production Function

- We will assume a café's only input is labor
 - This is clearly a simplifying assumption, cafes need capital like espresso machines as well

$$Y = f(L)$$

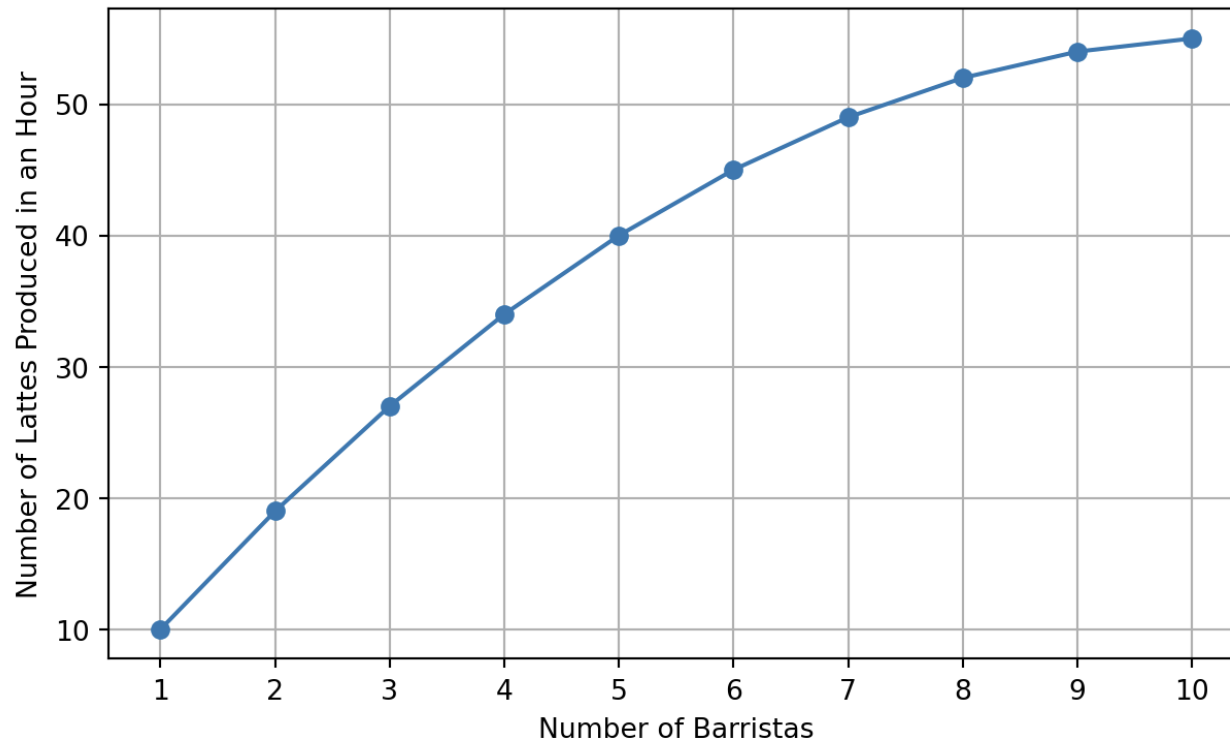
- To be concrete, let's consider a graphical illustration of a production function

The Production Function



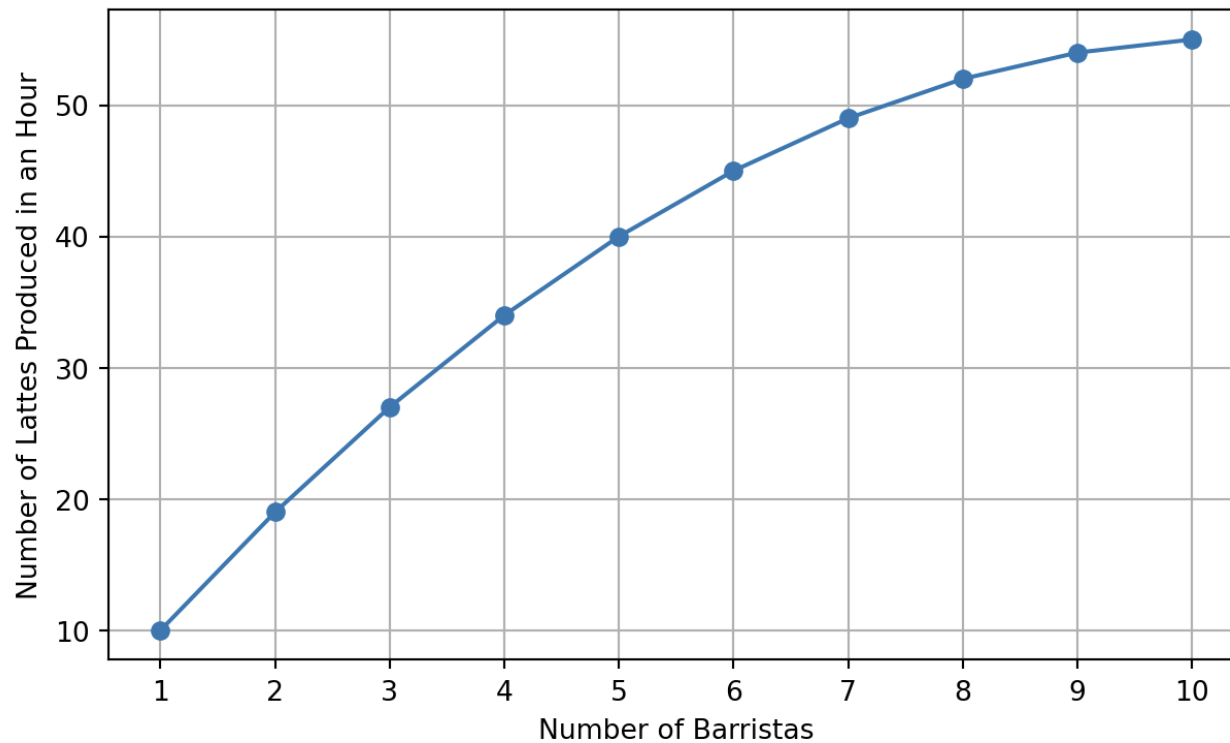
- Going from 0 to 1 barista increases number of lattes by 10

The Production Function



- Going from 0 to 1 barista increases number of lattes by 10
- Going from 1 to 2 baristas increases number of lattes by 9

The Production Function



- Going from 0 to 1 barista increases number of lattes by 10
- Going from 1 to 2 baristas increases number of lattes by 9
- This production function exhibits **decreasing marginal returns**

Diminishing Marginal Returns

- The more workers you hire, the lower the additional output of the next worker
 - Makes a lot of sense in the café. If you hire two baristas, you might be able to get through the line quicker and increase output, but there is still only one espresso machine
- The additional output you get from an extra worker is the **marginal product of labor (MPL)**

How many workers are hired?

- Let's consider a café that currently employs 1 worker? Should they hire another worker?
- Benefit: the marginal product of the second worker is 9. If you sell each latte for \$5 then the **marginal revenue product of labor (MRPL)** is \$45
- Cost: you need to pay the worker a wage. Let's assume the wage is \$15
- Benefit > cost ($45 > 15$), so you should hire the worker



How many workers are hired?

- Should we hire the 3rd worker?
- This will clearly depend on whether MRPL of the 3rd worker is > 15
- To save us some time, let's compute the MRPL for every additional worker

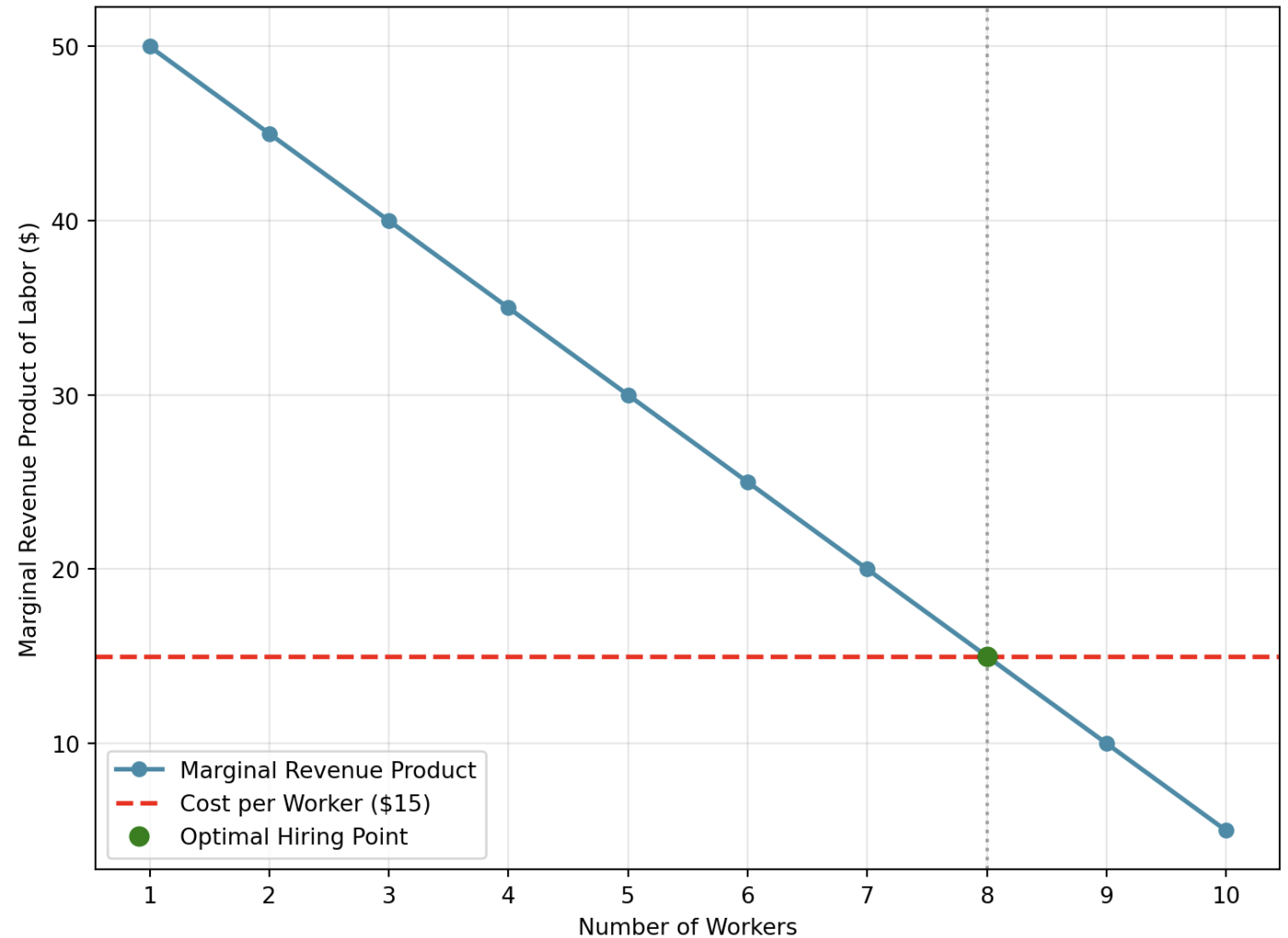
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Number of Workers	Marginal Product (Lattes)	Marginal Revenue Product (\$)
1	10	50
2	9	45
3	8	40
4	7	35
5	6	30
6	5	25
7	4	20
8	3	15
9	2	10
10	1	5

How many workers are hired?

- Keep hiring workers until the cost exceeds the MRPL
- In this example, this occurs at 8 workers



The Labor Demand Curve

- The labor demand curve is how many workers are hired at a given wage

$$L(w)$$

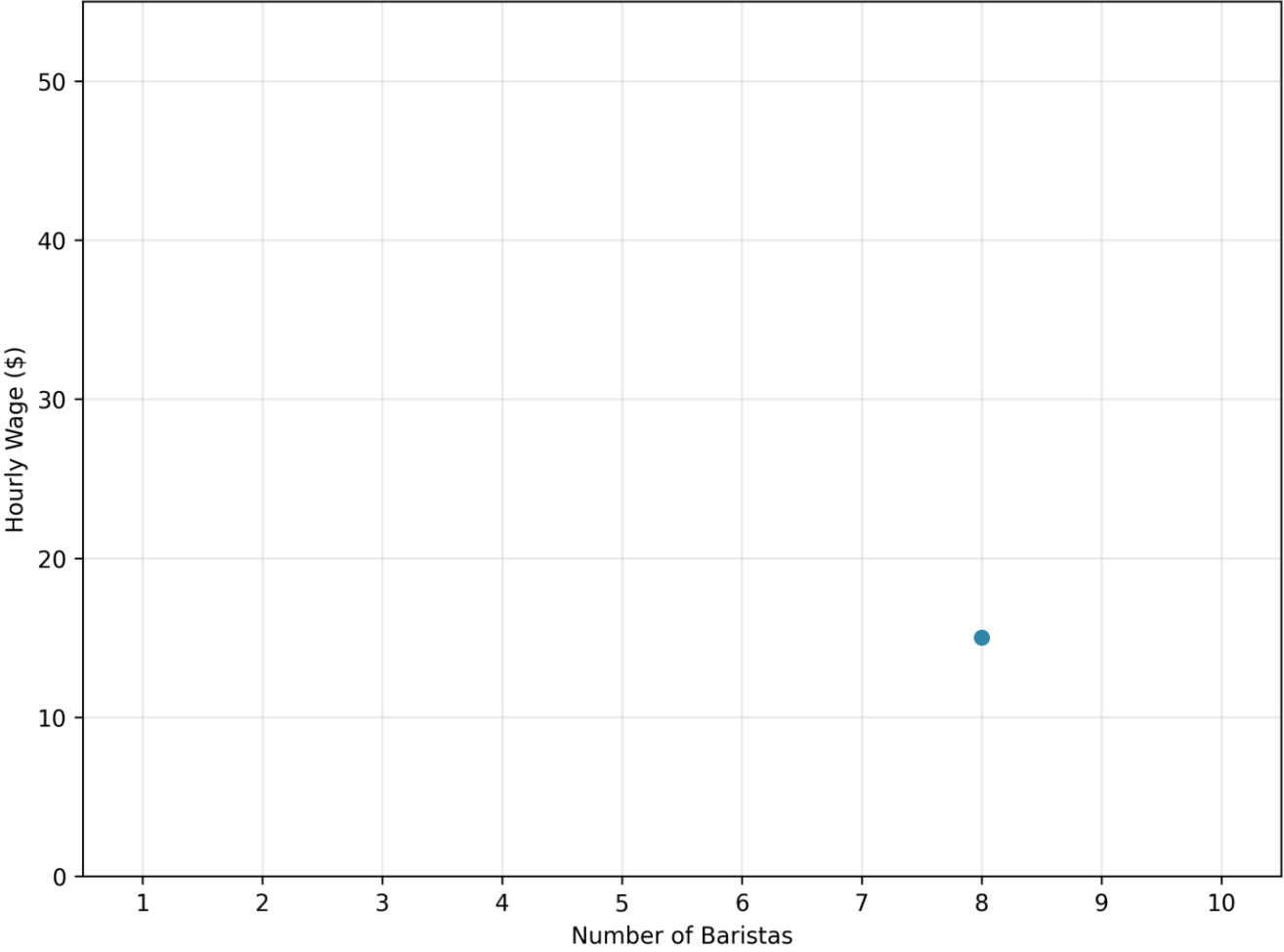
- We have found one point on the labor demand curve. At a wage of 15, the firm hires 8 workers

$$L(15) = 8$$

- To find the entire labor demand curve, we need to figure out how many workers are hired at all wages

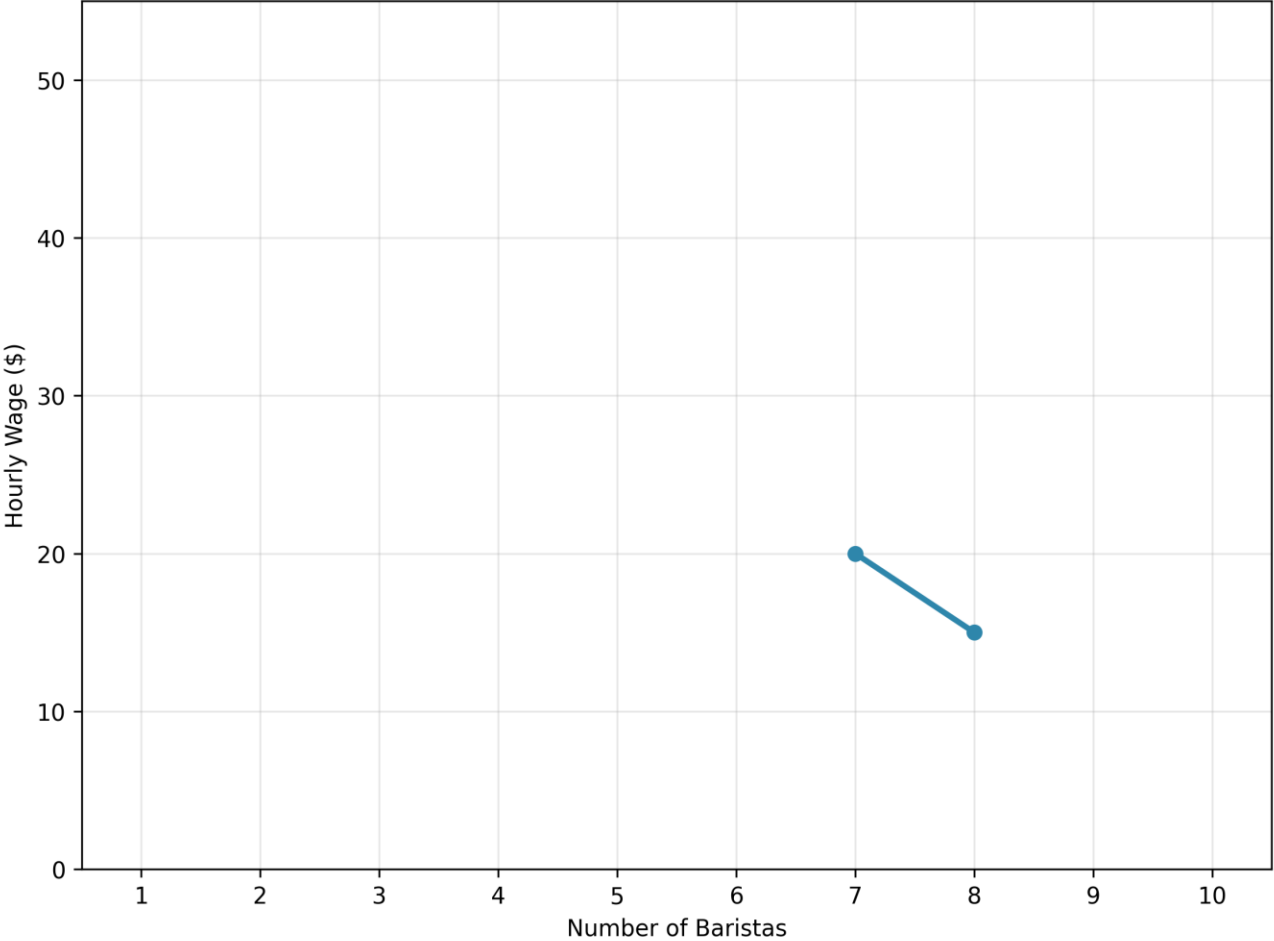
The (Inverse) Labor Demand Curve

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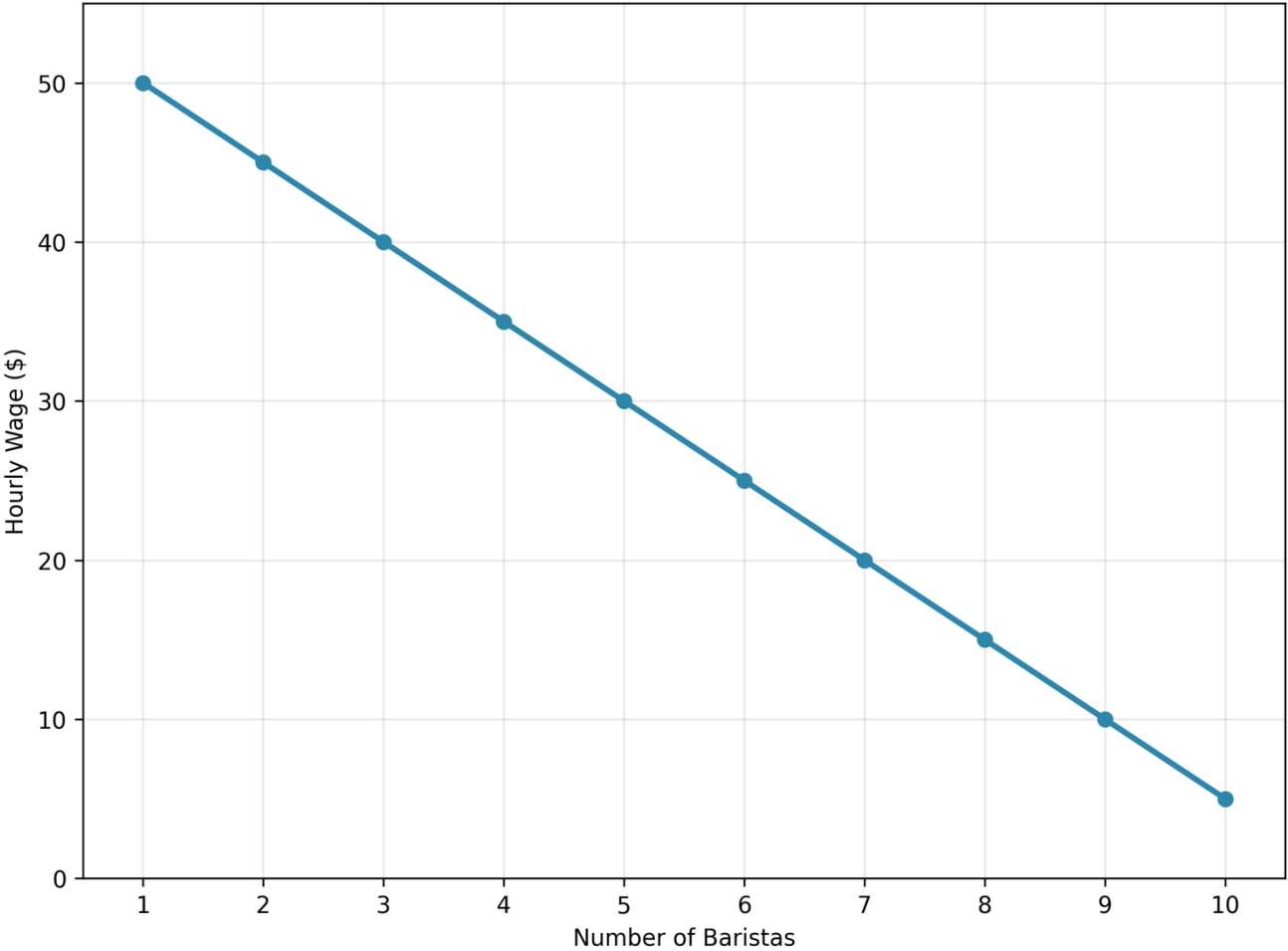
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The Labor Demand Curve

- You may have noticed that the labor demand curve is exactly the same as the MRPL curve
 - This is true in perfectly competitive models
- But conceptually, these two things are distinct and you should think about them as different concepts
 - Marginal Revenue Product of Labor is the value of hiring an additional worker
 - Labor demand curve is the amount of workers you hire for a given wage



Please submit your first and last name (as they appear on Canvas)



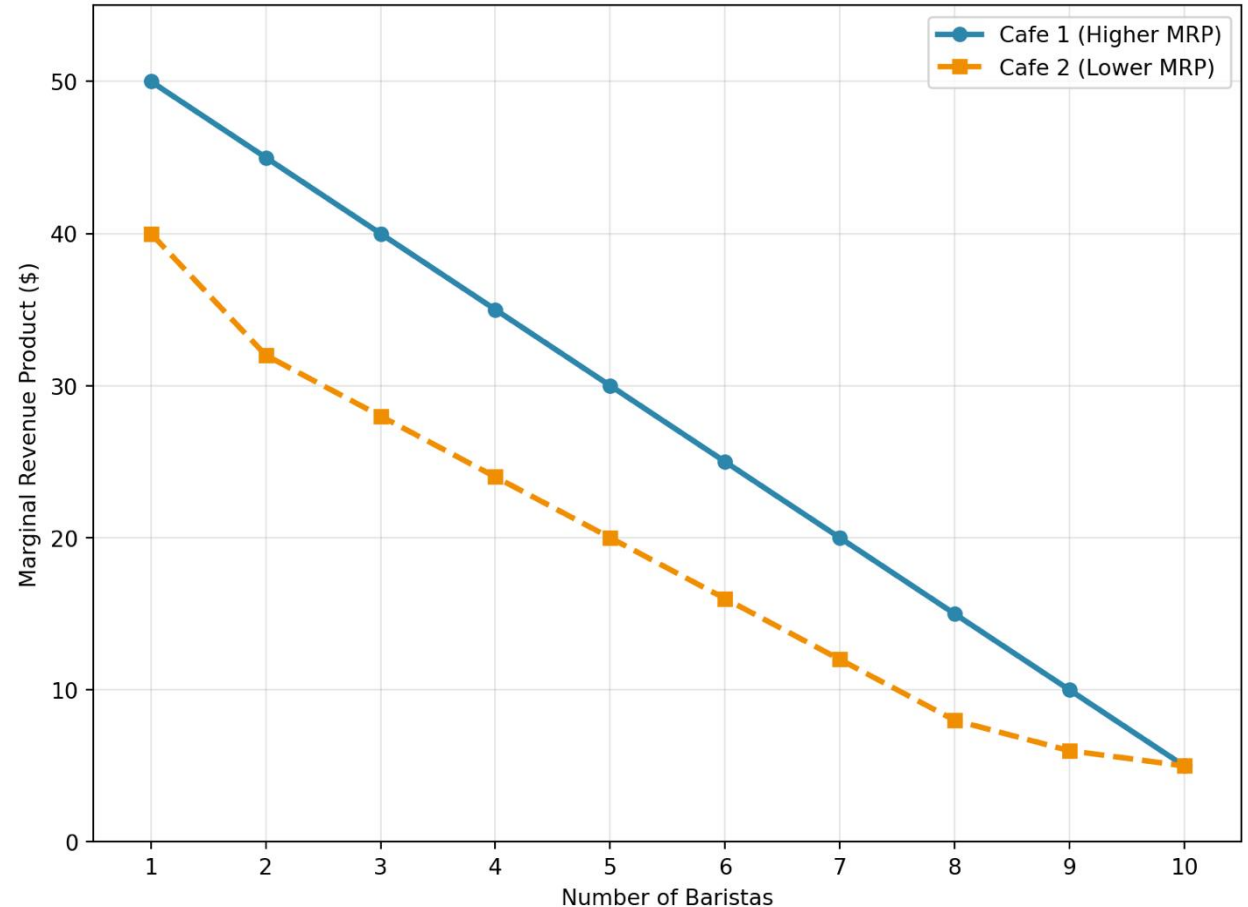
Imagine this cafe found a free espresso machine that increases the number of lattes per hour a worker can produce. Assuming the wage and price of lattes does not change, what happens to employment?

Market-Level Demand Curve

- So far we have figured out the labor demand curve for our café, but there are likely other firms in the market
- How do we form the market-level demand curve?
 - This is the curve that will be relevant for determining the market wage

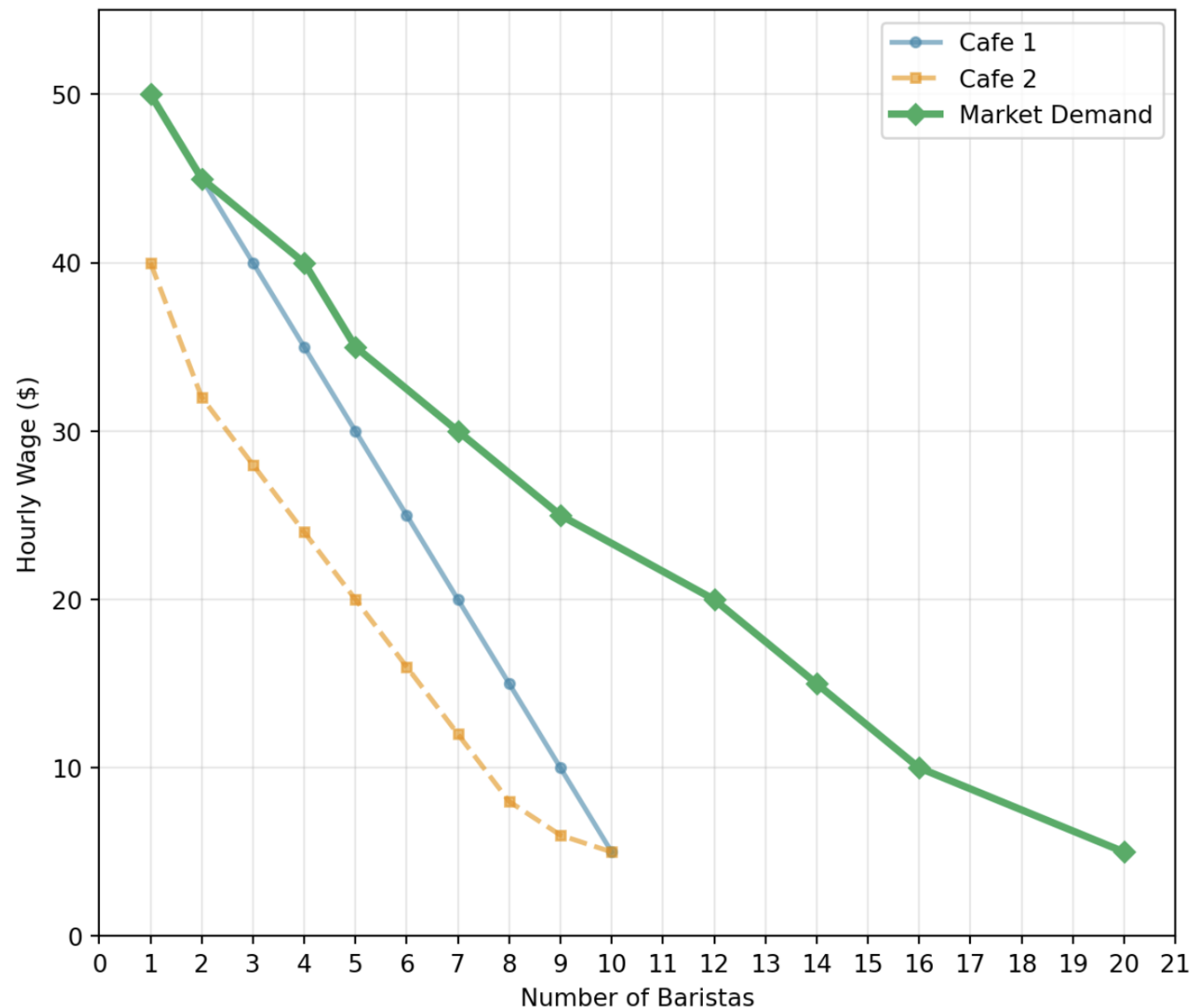
Market-level Demand Curve with 2 firms

- Let's start at a wage of 50. Café 1 hires 1 worker. Café 2 hires 0 workers, so market demand is 1
- At a wage of 40, café 1 hires 3 workers while café 2 hires 1, for a market-level demand of 4 workers
- We can continue this process on and on...



Market-level Demand Curve with 2 firms

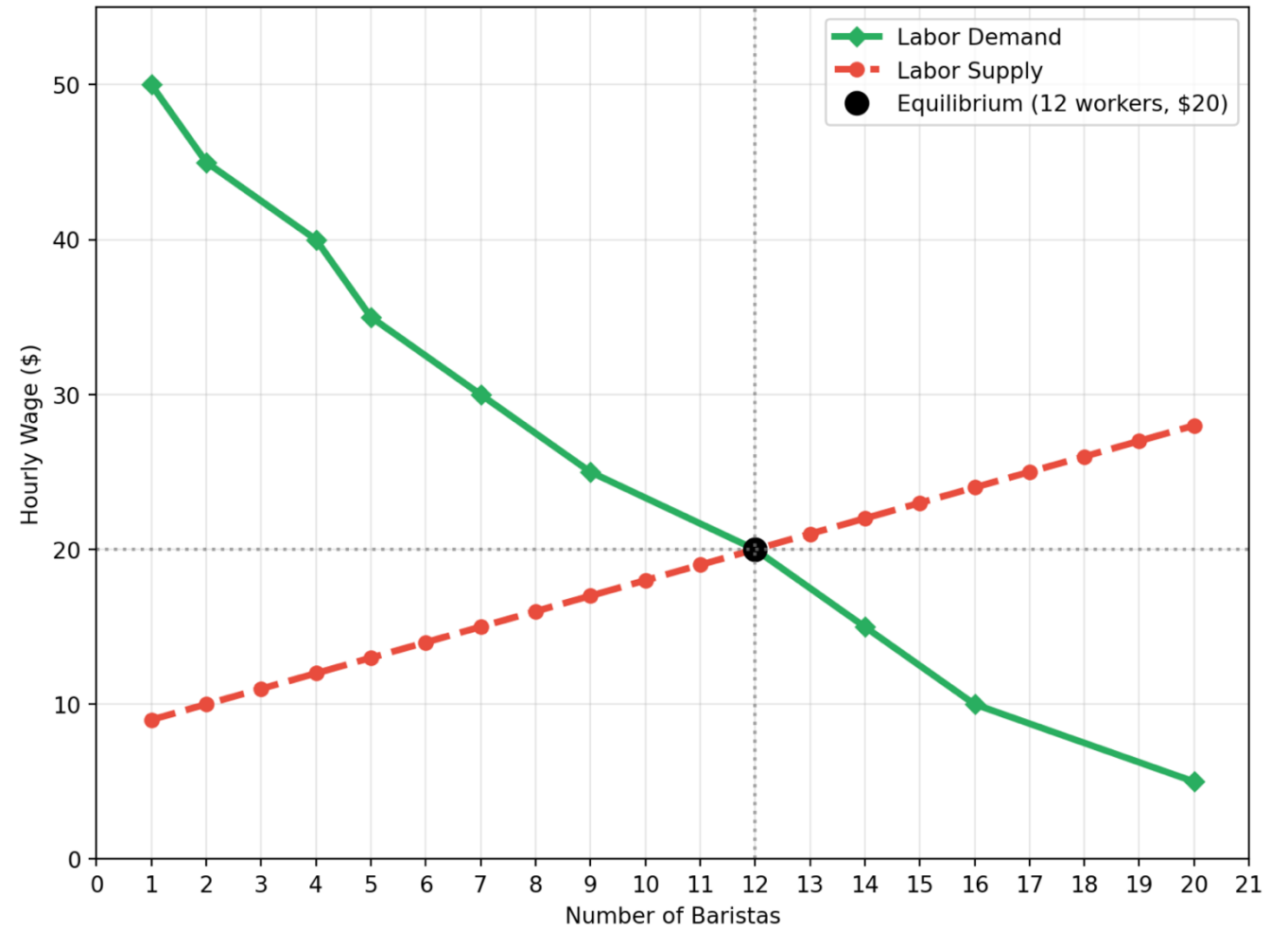
- With two cafes straightforward to compute market-level demand
- But exact same logic applies if there are 3, 50, 100, or 1000
- For each wage, sum up the demand of each individual firm



Labor Supply

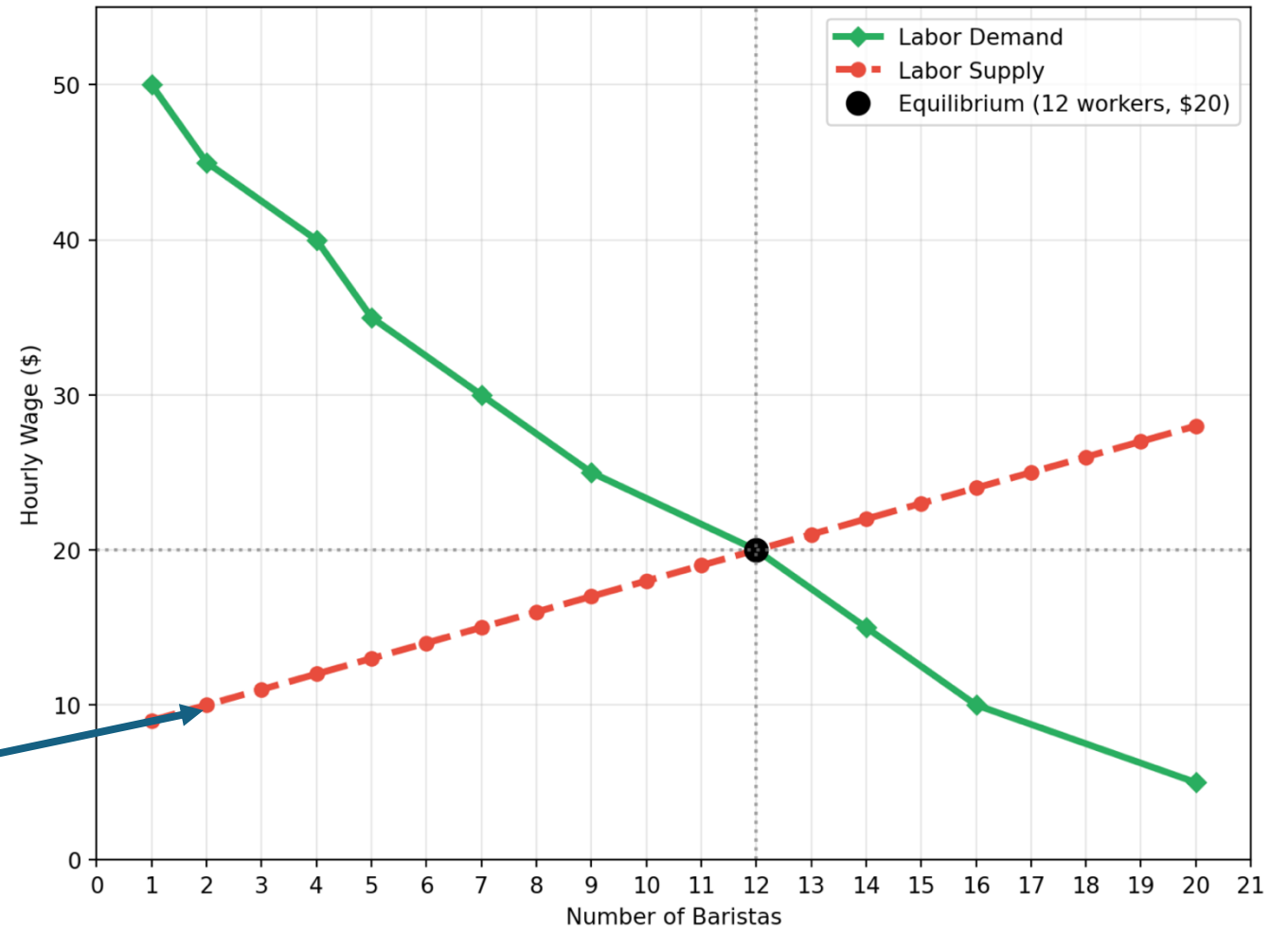
- We now have the market-level labor demand, but to figure out equilibrium employment and wages, we need the market-level labor supply
- Later in the course, we will go into detail about what determines the labor supply in a market, but for now we will assume it is upward sloping
 - In other words, more workers will want to work in this market if there is a higher wage. Seems like a reasonable assumption

Equilibrium



Equilibrium

Why isn't \$10 an equilibrium?

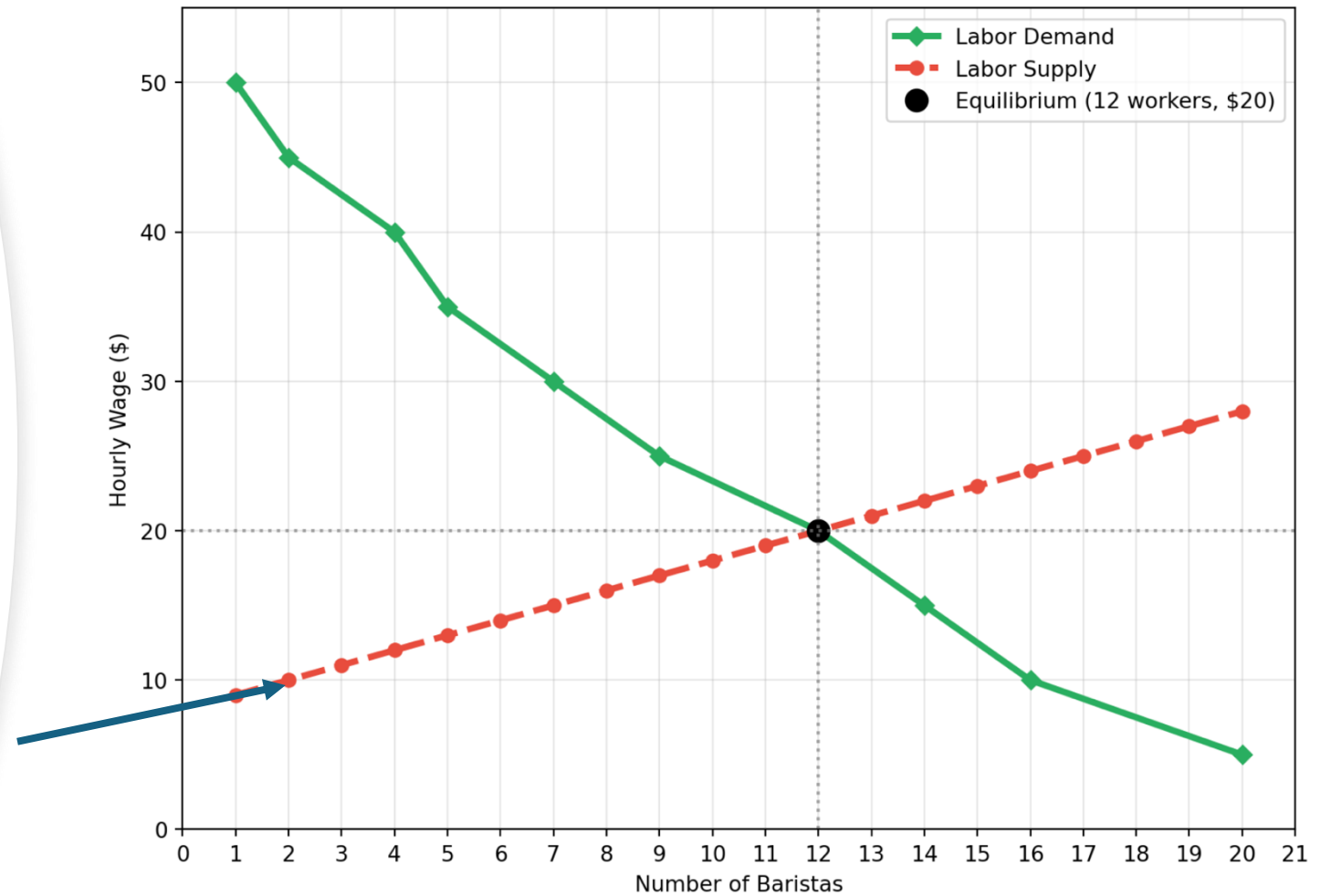


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Café 1 currently hires 1 worker
so makes $50 - 10 = 40$

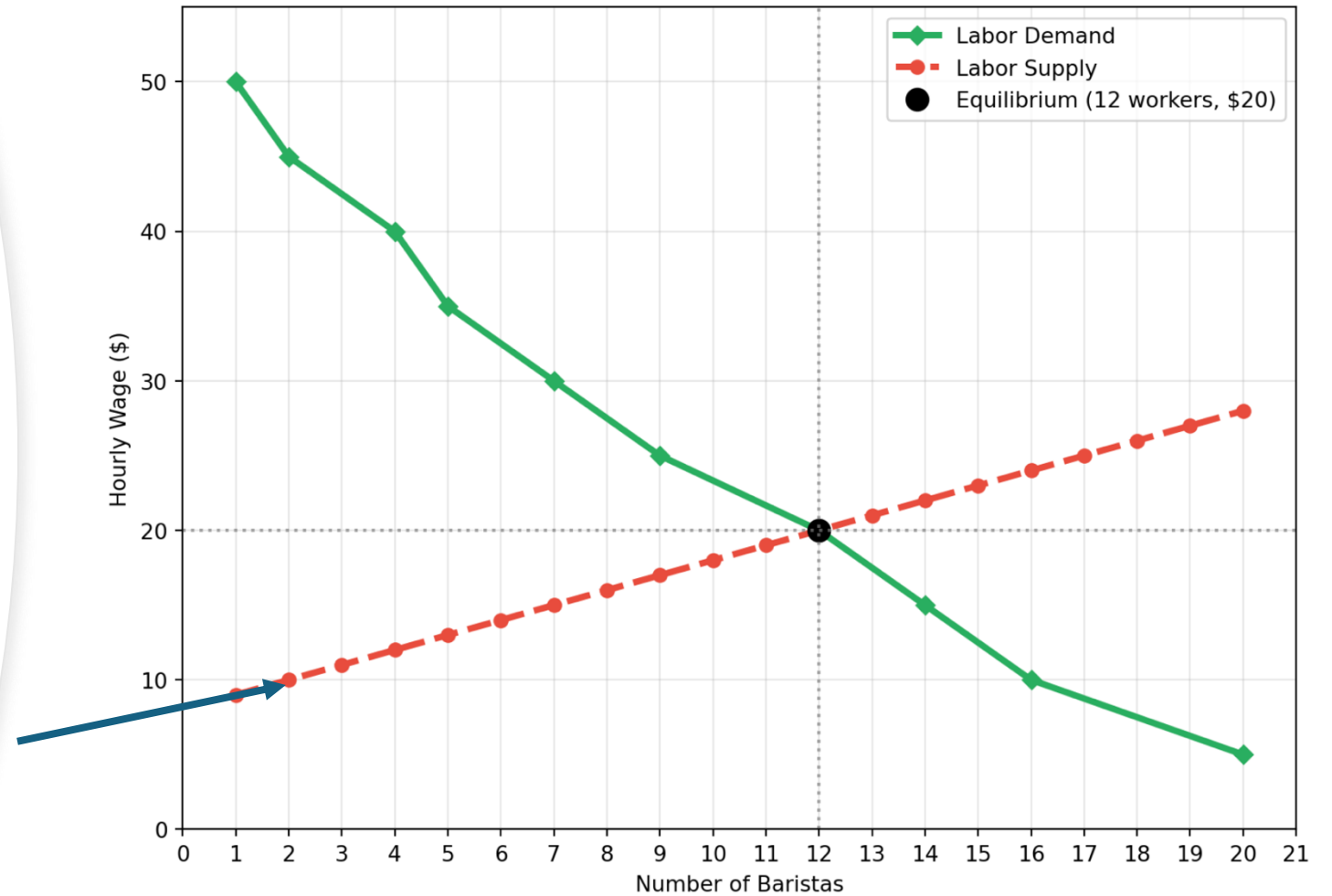


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If Café 1 raises wage to 11, can attract 3 workers and
produce 27 lattes for total profits of $27(5) - 3(11) = 102$



Imagine immigration increases the price of coffee by increasing demand, but also increases the supply of labor. The wage in the barrista labor market will?



Are there any concepts from today's lecture that are still unclear?

Concepts

- The key decision to hire is to compare **marginal revenue product of labor** against the cost of labor
- Doing this for every firm in the market yields a **market level** demand curve
- Combined with a market supply curve allows us to find the equilibrium level of employment and wages in a perfectly competitive labor market